AI-driven Virtual Lithography Experiments for Manufacturing Stability Optimization

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GPT-5.1, Gemini 3 Pro

Abstract

In semiconductor manufacturing, the lithography process plays a critical role and is highly sensitive to even minor parameter variations, which can lead to pattern defects and yield degradation. Moreover, in high-volume manufacturing environments, the high cost and long cycle time of physical experiments make it nearly impossible to exhaustively explore the entire lithography parameter space.

In this study, we propose an AI-based virtual lithography experimentation framework in which AI functions not merely as an analysis tool, but as a **primary and active experimenter**. Within a physics-informed virtual environment, the proposed AI autonomously designs virtual experiments through active learning, generates physically consistent synthetic process outcomes, and analyzes nonlinear correlations among lithography parameters, pattern defects, and yield-related metrics.

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1. Introduction

Lithography is one of the most critical, sensitive, and costly steps in the semiconductor manufacturing process. Variations in exposure dose, focus offset, and post-exposure bake (PEB) conditions directly affect critical dimension (CD) control, pattern integrity, and ultimately device yield.

As device scaling continues, the edge placement error (EPE) budget has significantly tightened. Consequently, preventing defects such as scumming and pattern collapse has become as important as conventional CD control. At the same time, the lithography process window is continuously shrinking, making stable process optimization increasingly challenging.

In real manufacturing environments, constraints such as cost, time, and equipment availability make it infeasible to physically test hundreds of parameter combinations in a production fab. Existing machine learning studies applied to lithography processes have largely focused on post hoc analysis of pre-collected datasets, failing to fully leverage the potential of AI to actively design experiments and explore the process space.

To address this limitation, this study proposes an **AI co-scientist framework** in which AI performs virtual lithography experiments, analyzes the results, and derives manufacturing-relevant insights. This approach prioritizes manufacturing stability and yield-oriented decision support over pure prediction accuracy.

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2. Related Work

AI applications in lithography research can be broadly categorized into three groups.

First, regression-based and deep learning models have been proposed to predict CD or overlay errors using lithography process parameters as inputs. While these approaches demonstrate high predictive accuracy, they typically assume the availability of large amounts of high-quality measured data.

Second, defect and pattern failure detection studies using SEM or inspection images have been widely reported. Although these methods achieve strong classification performance, they are often decoupled from process parameter optimization.

Third, explainable AI techniques such as SHAP have been applied to interpret the influence of lithography parameters. However, most of these studies remain at the visualization level and rarely translate into actionable items for real manufacturing environments.

Unlike prior work, this study positions AI as an **active virtual experimenter**, systematically exploring the virtual process space and directly deriving stable manufacturing windows.

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3. Problem Definition and Research Objective

Lithography processes involve complex and nonlinear interactions among multiple parameters, including exposure dose, focus offset, and PEB conditions, making it difficult to identify optimal process settings using limited experimental data alone. This study addresses the lack of scalable methods for exploring the lithography process space and deriving stable manufacturing conditions without physical experiments.

The objectives of this research are as follows:

- **Active Exploration:** Explore a wide parameter space through AI-driven virtual lithography experiments
- **Correlation Analysis:** Analyze correlations among lithography parameters, pattern defects, and yield-related metrics
- **Decision Support:** Derive stable lithography process windows that support manufacturing stability and decision-making

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4. AI-driven Virtual Experiment Framework

4.1 Virtual Experiment Design

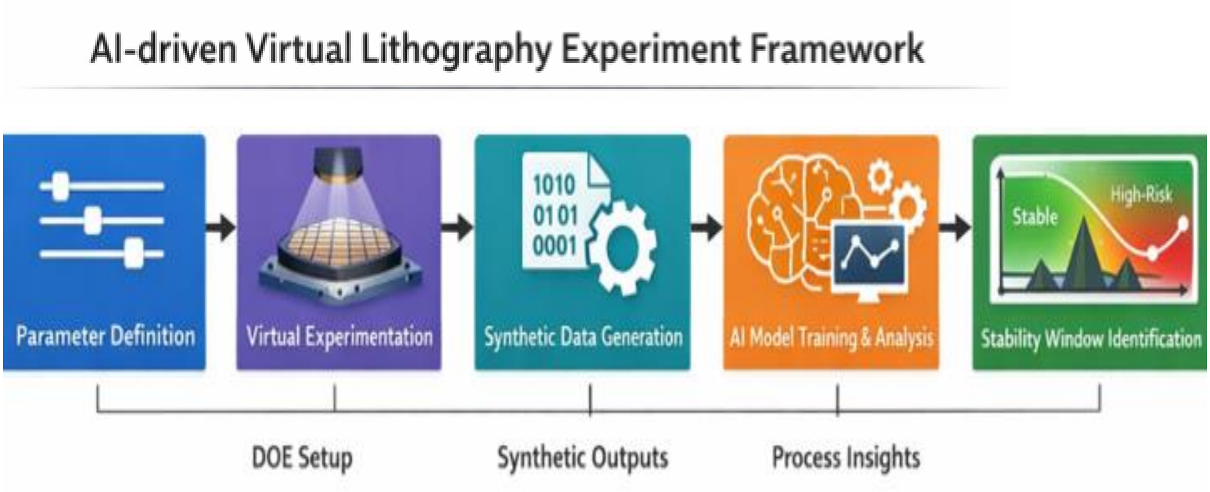
The AI defines manufacturing-relevant parameter ranges, including exposure dose, focus offset, PEB temperature, PEB time, and development time.
Design of Experiments (DOE)-based strategies are then used to efficiently generate parameter combinations for virtual experiments.

4.2 Synthetic Data Generation

For each virtual experiment, the AI generates outcomes such as ADI CD, ACI CD, CD bias, CD uniformity, and pattern defect probability.
Physical constraints and statistical variability are incorporated to ensure consistency with real manufacturing environments.

4.3 Model Training and Analysis

Machine learning models are trained to capture nonlinear relationships between process parameters and outcomes.
Explainable AI techniques are applied to interpret parameter importance and interaction effects



5. Virtual Dataset Description

The virtual dataset consists entirely of AI-generated synthetic data.
Each sample corresponds to a single virtual lithography experiment and includes associated process parameters and outcome metrics.

Key characteristics include:

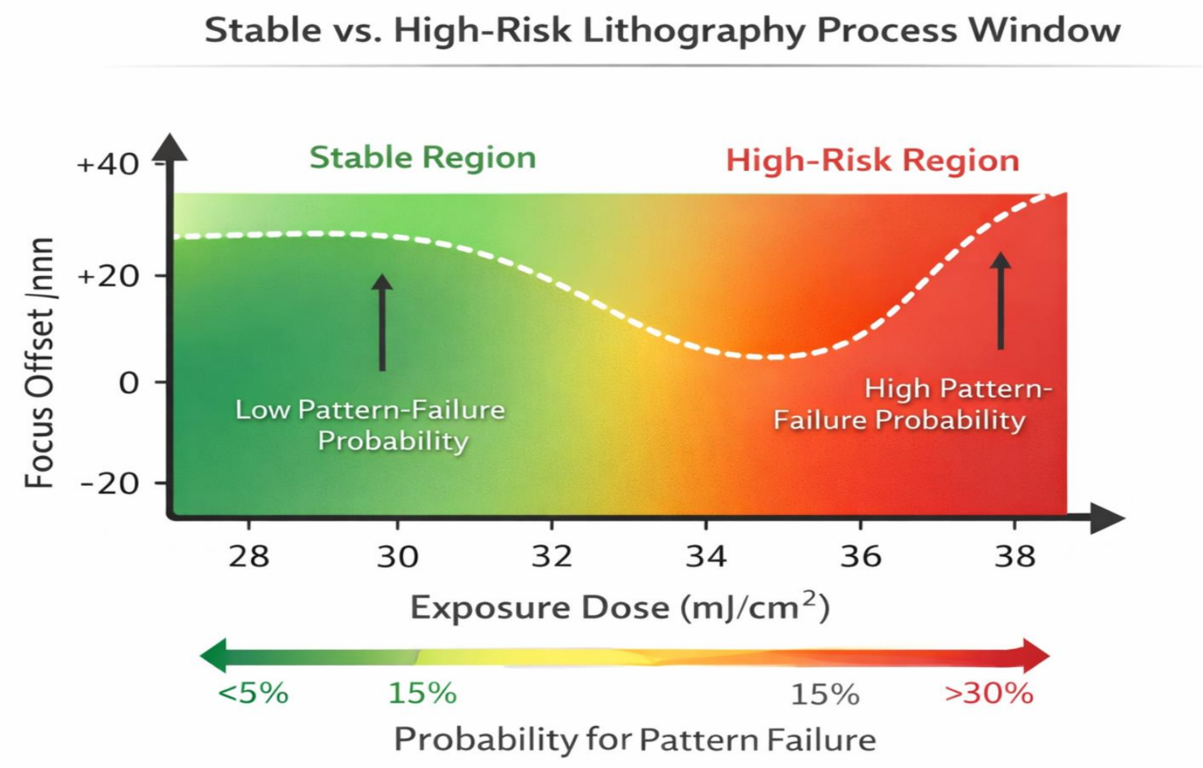
- Coverage of a high-dimensional parameter space (Dose, Focus, PEB Temperature/Time, Development Time)
- Inclusion of spec pass/fail labels as a proxy for yield
- Incorporation of random noise and variability observed in real processes

This enables comprehensive exploration of the lithography process space without physical experimentation.

6. Results and Analysis

The analysis reveals that focus offset is a dominant factor governing pattern defects.
Exposure dose is found to strongly interact with PEB conditions, significantly influencing CD uniformity.
Explainable AI analysis shows that defect risk increases nonlinearly once certain parameter thresholds are exceeded.
Based on these findings, a stable lithography process window with low defect probability

and acceptable CD uniformity is derived.



This illustrates the AI-identified stable and high-risk lithography process regions. The stable region corresponds to parameter combinations where pattern failure probability remains low despite moderate process variation.

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7. Manufacturing Implications

The proposed framework provides direct value to manufacturing and process engineers by enabling the derivation of stable process windows that support:

- Data-driven and robust process settings
- Margin-centered parameter selection that accounts for process variability
- Rapid re-optimization to prevent yield loss in the presence of process drift

The framework can be further extended to equipment-specific customization.

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8. Limitations and Future Work

As this study is based on synthetic data, it may not fully capture subtle differences

arising from real equipment behavior or photoresist chemistry.
Future work will apply transfer learning to calibrate the virtual model using a small amount of measured data and extend the framework to integrate additional process modules such as etching.

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9. Conclusion

This study presents a framework in which AI is utilized as a **co-scientist**, rather than a simple analysis tool, to systematically explore the lithography process space. The results suggest that AI-driven virtual experimentation can serve as a scalable, manufacturing-friendly approach for process optimization and yield stabilization.

AI Co-Scientist Challenge Checklist

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1. Does the study clearly define a scientific research question?

Yes
The study defines a clear research question focused on analyzing lithography process parameters, pattern failures, and manufacturing stability using AI-driven virtual experiments.

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2. Is AI used beyond simple data analysis or text generation?

Yes
AI acts as the primary experimenter by designing virtual lithography experiments, generating synthetic process data, analyzing results, and performing self-review.

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3. Does AI participate in experiment design?

Yes
AI defines manufacturing-relevant parameter ranges and constructs DOE-style virtual experiments to explore lithography process space.

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4. Does AI generate or acquire data?

Yes
AI generates physically consistent synthetic lithography data through virtual experimentation without relying on physical measurements.

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5. Does AI analyze the data and extract insights?

Yes
AI trains machine-learning models on synthetic data and applies explainable AI techniques to identify critical parameters and interactions affecting pattern failure and stability.

6. Does AI contribute to interpretation or decision making?

Yes
AI derives stable lithography process windows and distinguishes stable versus high-risk regions to support manufacturing-oriented decision making.

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7. Is explainable or interpretable AI used?

Yes
Explainable AI methods (e.g., SHAP-style interpretation) are applied to quantify parameter contributions and interaction effects.

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8. Does the study demonstrate AI as a “co-scientist” rather than a tool?

Yes
AI performs experiment design, execution, analysis, and self-review, while the human researcher supervises and validates physical plausibility.

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9. Is the role of the human researcher clearly defined?

Yes
The human researcher defines the research problem, validates AI-generated results, ensures manufacturing relevance, and draws final conclusions.

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10. Are limitations of AI involvement explicitly discussed?

Yes
The study acknowledges reliance on synthetic data and absence of physical tool validation, and discusses these as limitations.

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11. Does the study suggest future improvements or extensions?

Yes

Future work includes calibration with real measurement data, tool drift modeling, and extension to other semiconductor process modules.

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12. Is the use of AI ethically and transparently disclosed?

Yes
All AI contributions, including virtual experimentation, data generation, analysis, and writing assistance, are transparently documented.

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13. Are the results relevant to real-world scientific or engineering problems?

Yes
The study addresses lithography process stability and yield-related decision making, which are critical challenges in semiconductor manufacturing.

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14. Is the study reproducible in principle?

Yes
The virtual experiment framework, parameter definitions, and analysis flow are fully described, enabling conceptual reproducibility.

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15. Does the work align with the goals of the AI Co-Scientist Challenge?

Yes
The study demonstrates AI as an active scientific collaborator throughout the research lifecycle, aligning directly with the challenge objectives.